Assignment 6

Technical Analysis using R, Visualization Phase

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Course: BDA400 - Data Science Tools and Techniques

Assignment title: Assignment 6 - Technical Analysis using R, Visualization Phase

Project description

This is the third stage of the three-part Technical Analysis project. It is an interactive R Shiny portfolio dashboard that downloads historical stock data (Yahoo Finance through quantmod, with an automatic Stooq fallback) and displays it as a line graph, candlestick or area chart on a Daily, Weekly or Monthly time frame. Moving Averages (SMA/EMA), RSI and MACD can each be switched on or off as extra layers on the same chart, with adjustable parameters. Four trading rules (Moving Average Crossover, MA Crossover with RSI filter, MACD Signal-Line Crossover, RSI Overbought/Oversold) generate Buy, Sell and Hold signals, which are annotated on the chart and listed in a table that can be downloaded as CSV.

Files in the repository

- SodiqYunusa_BDA400_A06.R - R script with all code, organised in sections 1 to 5 matching the assignment tasks.
- Portfolio.txt
- SodiqYunusa_BDA400_A06_CoverPage.pdf- this cover page.

Repository link

https://github.com/SodiqYunusa/TechnicalAnalysis/tree/main/Assignment6